

WORLD DEVELOPMENT STATISTICS

Targeted Funding for "Hollow Growth" Regions

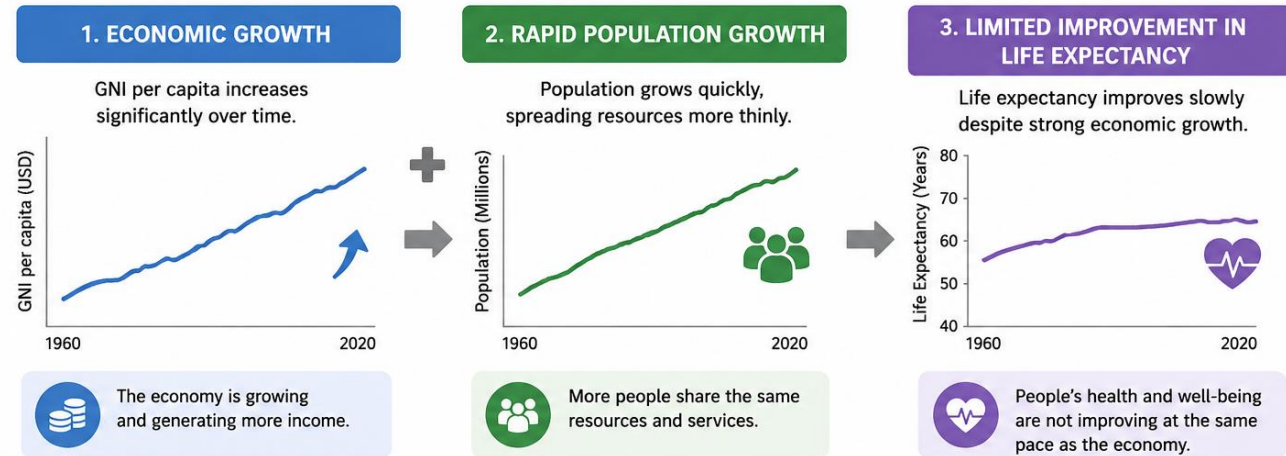
OUTLINE

- Introduction
- Problem statement
- Data preparation methodology
- Analyzing data
- Identify Hollow-Growth Countries and validating results
- Recommendation and Summary

WHAT IS HOLLOW GROWTH REGIONS?

HOLLOW GROWTH: STRONG ECONOMY, WEAK HUMAN DEVELOPMENT

Economic growth does not always translate into better lives. When population grows rapidly, the benefits of rising income may not reach people, resulting in limited improvements in life expectancy.





HOLLOW GROWTH

High economic growth + Rapid population growth
but Low improvement in life expectancy.



TARGETED FUNDING NEEDED

Identify and support countries experiencing
hollow growth to ensure economic gains
translate into better lives.



This project analyzes data from 1960–2020 to identify countries where economic growth
has not led to comparable improvements in life expectancy.

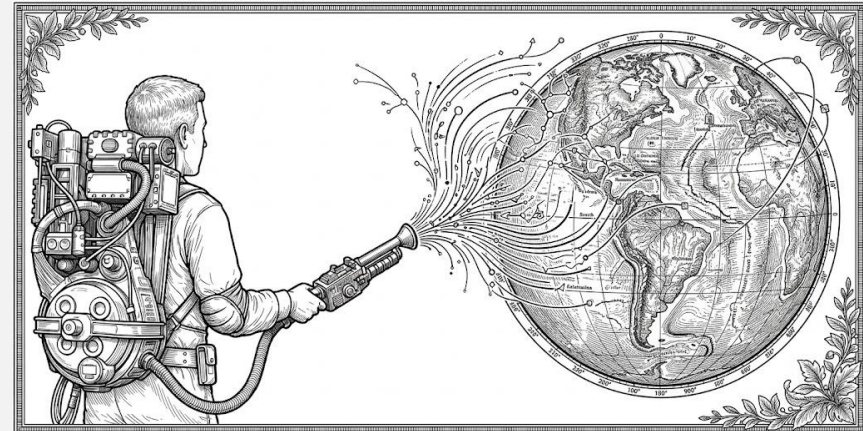


DATA PREPARATION METHODOLOGY

Three Gapminder World Development Statistics datasets were used:

- Population
- GNI per Capita in USD
- Life Expectancy

It covers 197 countries with current and predicted futuristic data
ranged from year 1800 → 2100.



To prevent getting biased results and to ensure fair comparison, seven countries were dropped from the datasets due lack in required data:

San Marino

North Korea

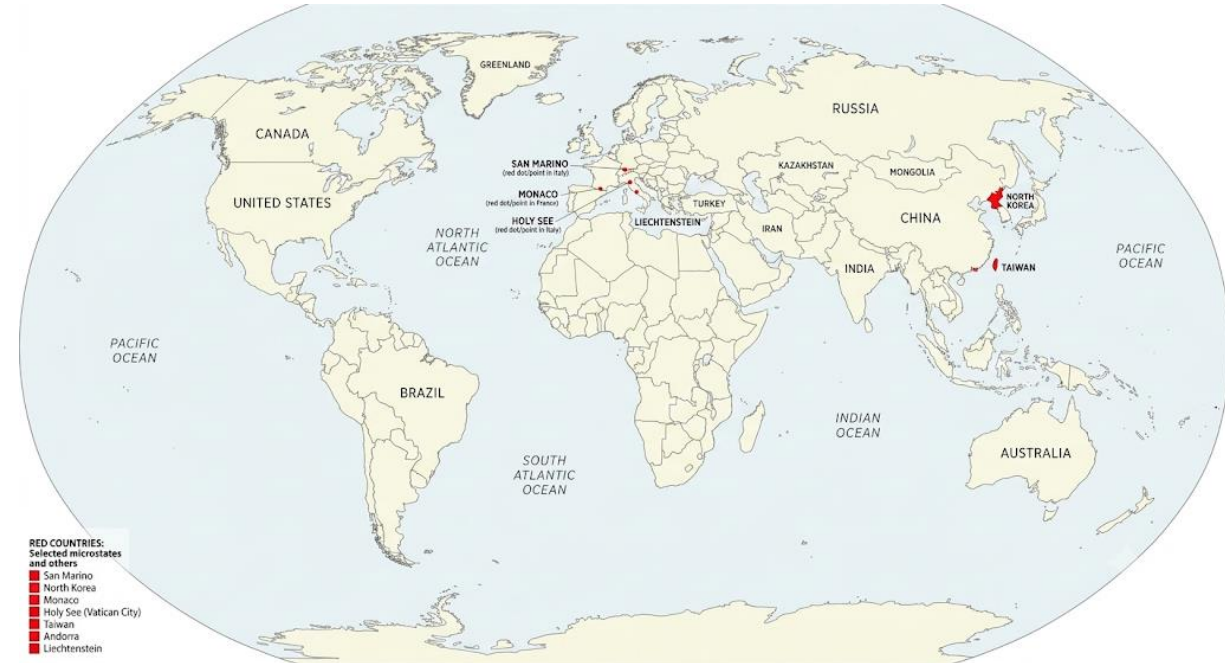
Monaco

Holy See

Taiwan

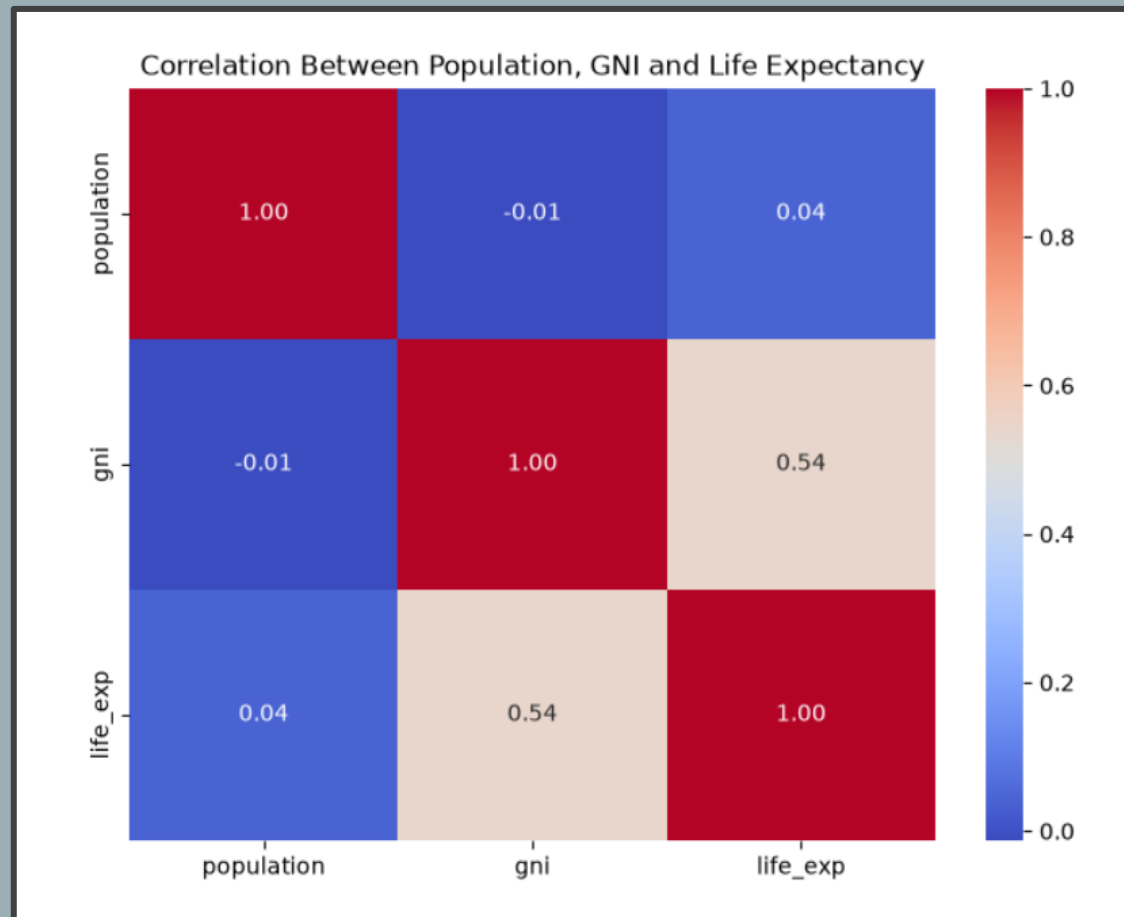
Andorra

Liechtenstein



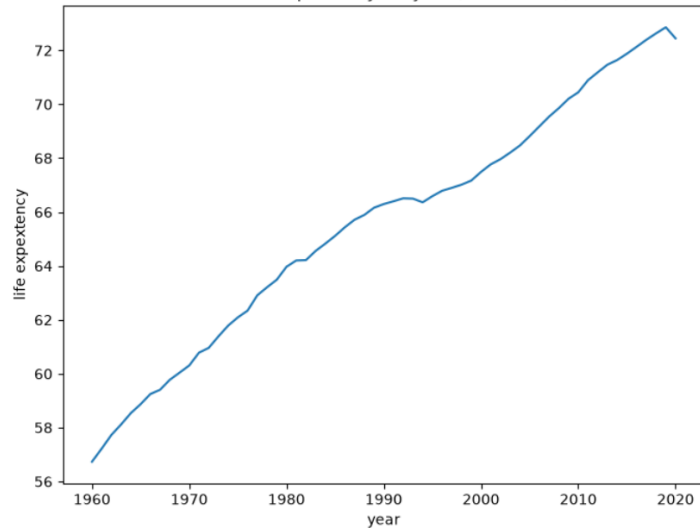
Around 80% of the overall data was retained.

ANALYZING DATA

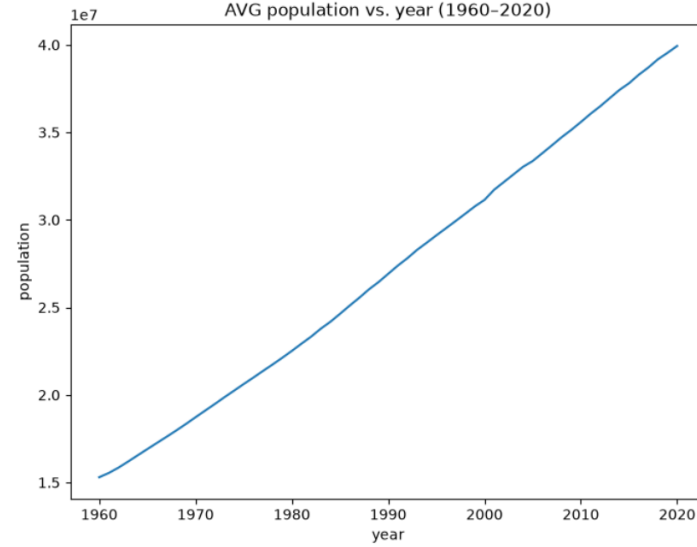


ANALYZING DATA

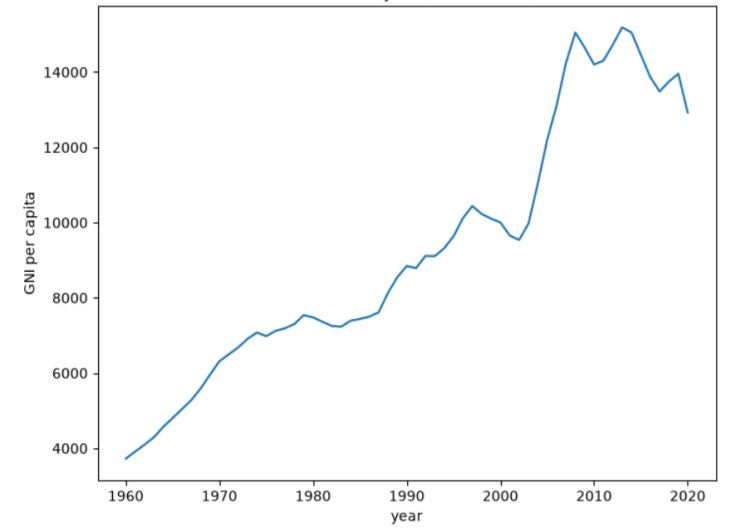
AVG life expectancy vs. year (1960-2020)



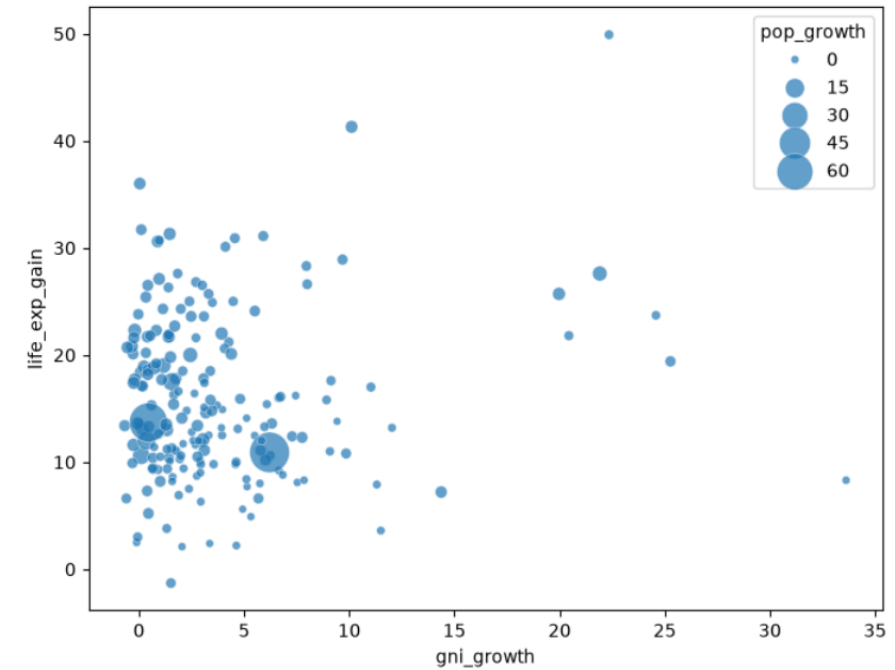
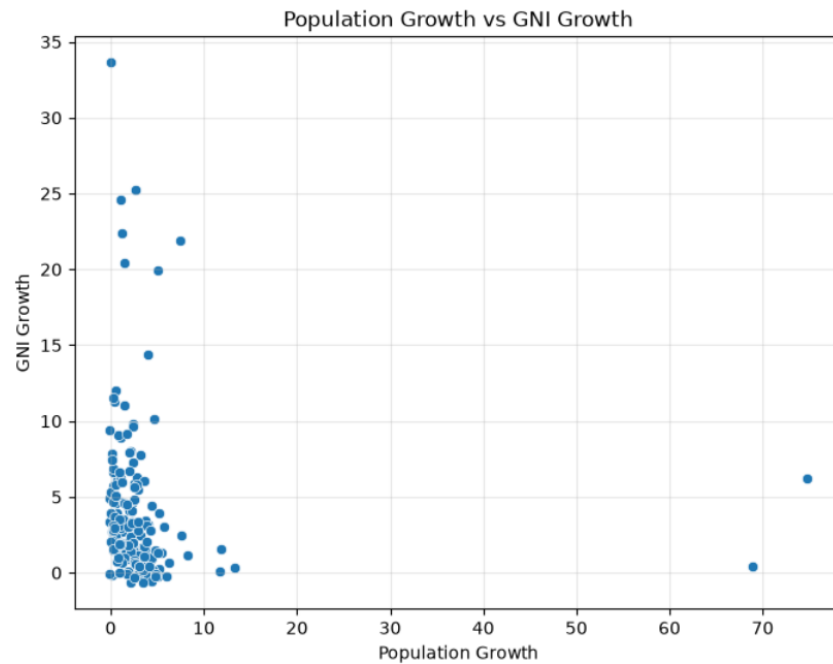
AVG population vs. year (1960-2020)



AVG GNI vs. year (1960-2020)



ANALYZING DATA



Top 5 GNI per capita growth rate countries			
country	pop_growth	life_exp_gain	gni_growth
Romania	0.05	8.3 years	33.62
Singapore	2.69	19.4 years	25.27
South Korea	1.01	23.7 years	24.58
China	1.17	49.9 years	22.35
Oman	7.45	27.6 years	21.91

Top 5 life expectancy gain countries			
country	pop_growth	life_exp_gain	gni_growth
China	1.17	49.9 years	22.35
Maldives	4.61	41.3 years	10.11
Cameroon	4.18	36 years	0.04
Iran	3.08	31.7 years	0.11
Yemen	4.83	31.3 years	1.46

Top 5 population growth rate countries			
country	pop_growth	life_exp_gain	gni_growth
Qatar	74.82	10.9 years	6.21
United Arab Emirates	68.85	13.7 years	0.45
Kuwait	13.3	12 years	0.36
Jordan	11.85	17.5 years	1.55
Djibouti	11.69	10.6 years	0.11

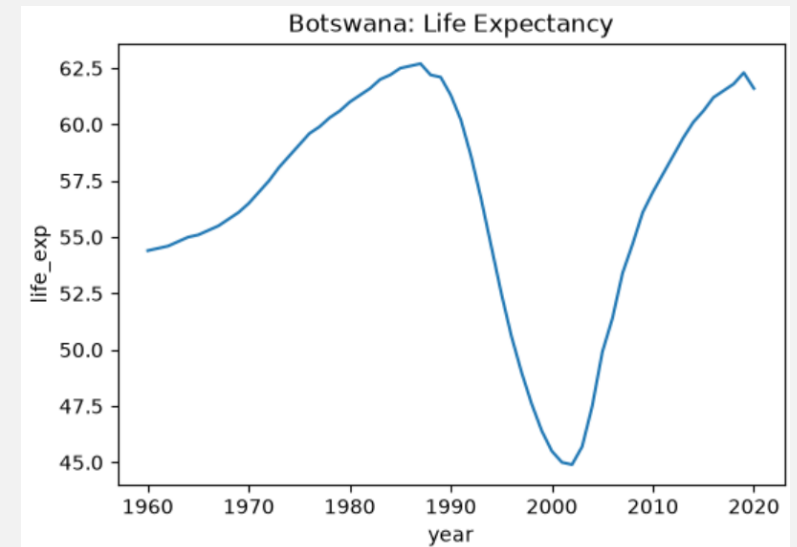
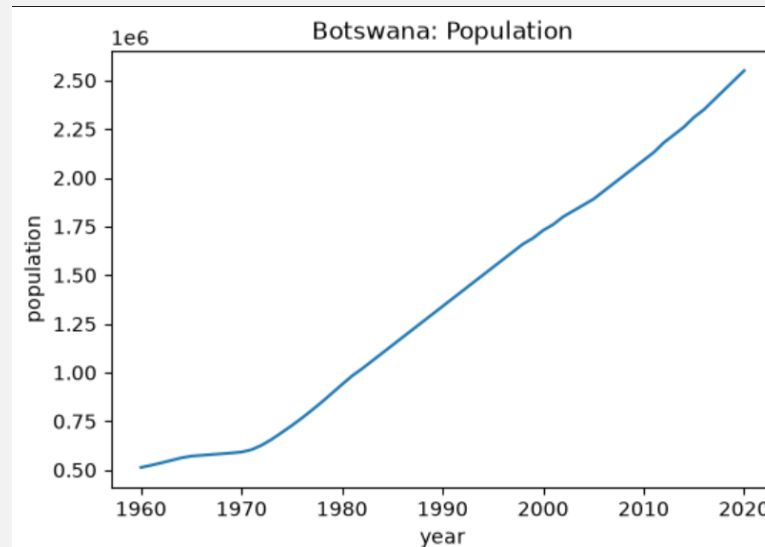
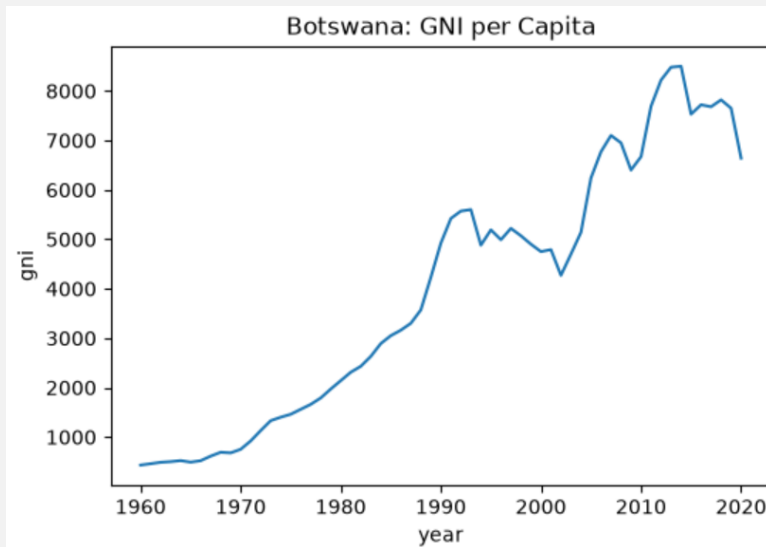
STATISTIC TABLE

	GNI_GROWTH	POP_GROWTH	LIFE_EXP_GAIN
count	190.000000	190.000000	190.000000
mean	3.830053	3.096053	15.787947
std	4.960800	7.426849	7.727380
min	-0.680000	-0.160000	-1.300000
25%	0.937500	0.680000	10.425000
50%	2.495000	2.125000	13.950000
75%	4.900000	3.507500	20.675000
max	33.620000	74.820000	49.900000

RESULTS

Hollow Growth countries

country	population_1960	population_2020	life_exp_1960	life_exp_2020	gni_1960	gni_2020	pop_growth	life_exp_gain	gni_growth
Botswana	513000	2550000	54.4 years	61.6 years	432	6640	3.97	7.2 years	14.37



RECOMMENDATION

- Based on combining historical trend analysis, correlation analysis and the data filtering methodology, it shows that Botswana is the strongest candidate for the target development fund.



Botswana Flag



SUMMARY

High Economic growth does is not always translated into better lives. By combining historical trend analysis, correlation analysis, and applying filtration methodology, the project successfully highlighted potential **hollow-growth** countries, which in our case was Botswana. These findings provide the Policy Steering Committee with evidence-based candidates for the proposed **\$100 million intervention fund**, ensuring that resources are directed toward countries experiencing stalled human development despite economic progress.